INTERNATIONAL SEARCH REPORT

International application No.
PCT/US 22/21289

A. CLASSIFICATION OF SUBJECT MATTER

IPC - INV. G01N 33/68, G01N 21/65, G16B 15/00, G16C 20/20; ADD. C07K 14/00 (2022.01)

CPC - INV. G01N 33/68, G01N 21/65, G16B 15/00, G16C 20/20, C12P 21/005, G01N 2400/10, G01N 2440/38; ADD. C07K 14/00

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
See Search History document

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched
See Search History document

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
See Search History document

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	BREWSTER et al. "Monitoring the Glycosylation Status of Proteins Using Raman Spectroscopy" Anal. Chem. 2011, Vol. 83, No. 15, pp 6074-6081, especially, Abstract; page 6075, Col. 2, para 3; page 6076, Col. 2, para 3; page 6078, Col. 1, para 4; page 6078, Col. 2, para 1-2; page 6080, Col. 1, para 3-4	1-4
Y	US 2017/0355947 A9 (BIOGEN MA INC.) 14 December 2017 (14.12.2017) Abstract; para [0004]; para [0056]; para [0214]; para [0216]; claim 1; claim 9	1-4
A	US 2014/0349337 A1 (MASSACHUSETTS INSTITUTE OF TECHNOLOGY) 27 November 2014 (27.11.2014), whole document	1-4

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
10 June 2022	AUG 15 2022
Name and mailing address of the ISA/US Mail Stop PCT, Attn: ISA/US, Commissioner for Patents P.O. Box 1450, Alexandria, Virginia 22313-1450 Facsimile No. 571-273-8300	Authorized officer Kari Rodriquez Telephone No. PCT Helpdesk: 571-272-4300

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Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☒ Claims Nos.: 5-15, 19-25, 29-37, 41-43
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:
This application contains the following inventions or groups of inventions which are not so linked as to form a single general inventive concept under PCT Rule 13.1. In order for all inventions to be searched, the appropriate additional search fees must be paid.

Group I: Claims 1-4 directed to a method for determining a glycan structure on a glycosylated molecule.

Group II: Claims 16-18, 26-28 and 38-40 directed to a method or system of producing a glycosylated molecule.

The inventions listed as Groups I and II do not relate to a single general inventive concept under PCT Rule 13.1 because, under PCT Rule 13.2, they lack the same or corresponding special technical features for the following reasons:

Special Technical Features

----------continued on supplemental sheet----------

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:
1-4

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

Box III: Observations where unity of invention is lacking:

Group I requires a method for determining a glycan structure on a glycosylated molecule, first bioreactors having a first volume equal to or below a first threshold, one or more second bioreactors having a second volume equal to or above a second threshold to result in measured spectral data; and determining, by at least one computing device using the generated one or more regression models and based on the measured spectral data, levels of the one or more glycan structures on the glycosylated molecule within one or more second bioreactors, not required by Group II.

Group II requires producing one or more glycosylated molecules comprising means for culturing a glycosylated molecule-producing cell line; means for measuring a level of one or more glycan structures and means for measuring a level of one or more glycan structures in a glycosylated molecule producing cell line, not required by Group I.

Common Technical Features

The features shared by the inventions listed as Group I and II are a glycan structure on a glycosylated molecule, obtaining, for each of a plurality of runs, levels for one or more glycan structures on the glycosylated molecule using a process analytical technology (PAT) tool, one or more regression models based on the obtained spectral data, determining, by at least one computing device.

However, these shared technical features do not represent a contribution over prior art, because the shared technical features are taught by an article entitled "Monitoring the Glycosylation Status of Proteins Using Raman Spectroscopy" by Brewster et al. (Anal. Chem. 2011, Vol. 83, No. 15, pp 6074-6081) (hereinafter 'Brewster').

Brewster discloses a glycan structure on a glycosylated molecule, obtaining, for each of a plurality of runs, levels for one or more glycan structures on the glycosylated molecule using a process analytical technology (PAT) tool, one or more regression models based on the obtained spectral data, determining, by at least one computing device (Abstract- "the characterization of these biopharmaceuticals poses a significant analytical challenge. In particular, monitoring posttranslational modifications (PTMs), such as glycosylation, is an important aspect of this characterization because these glycans can strongly affect the stability, immunogenicity, and pharmacokinetics of these biotherapeutic drugs. Raman spectroscopy is a powerful tool, with many emerging applications in the bioprocessing arena. Although the technique has a relatively rich history in protein science, only recently has Raman spectroscopy been investigated for assessing posttranslational modifications, including phosphorylation, acetylation, trimethylation, and ubiquitination. In this investigation, we develop for the first time Raman spectroscopy combined with multivariate data analyses, including principal components analysis and partial least-squares regression, for the determination of the glycosylation status of proteins"; page 6076, Col. 1, para 2; page 6076, Col. 2, para 1-3).

As the technical features were known in the art at the time of the invention, they cannot be considered special technical features that would otherwise unify the groups.

Groups I and II therefore lack unity of invention under PCT Rule 13 because they do not share a same or corresponding special technical feature.

Item 4 (continued)
Claims 5-15, 19-25, 29-37 and 41-43 are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).